TOPIC 5 · COPILOT AGENTS AND MINI CAPSTONE
LAB 11 THEME · COPILOT AGENT BUILDER

Lab 11 — Build a Work-Order Agent with Agent Builder

Maps to	LO5: build a grounded Copilot agent and apply the full workflow in a group capstone
Duration	30 minutes
Tools	Microsoft Copilot (copilot.cloud.microsoft) · Agent Builder · OneDrive
Mock data	data/knowledge/Northstar_Service_Handbook.docx, data/knowledge/Northstar_Work_Orders_Clean.xlsx
Course	GenAI for Engineers · v2.3
Prompts	https://tertiarycourses.github.io/IES-GenAI-for-Engineers/prompts/#lab-11

Goal

A private Northstar Work Order Assistant agent with reviewed instructions, two OneDrive knowledge files, the work-order-qa Skill (or its labelled manual route), four starter prompts and a verified test record.

What You Will Do

Create Northstar Work Order Assistant, a specialist Copilot agent that answers questions about the 30 September work-order snapshot. Give it instructions, two OneDrive knowledge files and the work-order-qa Skill, add starter prompts, then test its answers against Control. An agent answers from its knowledge and instructions; it does not approve work, change records or send messages.

What You Will Build

A private Northstar Work Order Assistant agent with reviewed instructions, two OneDrive knowledge files, the work-order-qa Skill (or its labelled manual route), four starter prompts and a verified test record.

Prerequisites

- Microsoft 365 account with Copilot available in Word, Excel, PowerPoint and Outlook. Availability depends on the assigned licence and tenant settings; confirm access with the trainer before class.
- Desktop Excel on Windows or Mac is recommended; save working copies to your assigned OneDrive course folder with AutoSave on. Do not overwrite the supplied input.
- Download this whole lab folder. All files are synthetic; work-genai is your own learner output folder.
- A Microsoft 365 Copilot licence (or pay-as-you-go) with Agent Builder enabled by your administrator. Skills in agents are a preview for Frontier-enrolled tenants; otherwise use the labelled manual route.

Data note

Northstar Water Services is a fictional water-pipe maintenance and service company. All service hubs, staff and work orders are synthetic. Use the fixed reporting date 2026-09-30; missing facts remain UNKNOWN. Proposals do not change real assignments. Training email drafts are never sent.

Steps

Step 1. Create a OneDrive folder work-genai/agent-knowledge and upload data/knowledge/Northstar_Service_Handbook.docx and data/knowledge/Northstar_Work_Orders_Clean.xlsx. The workbook has a single sheet so the agent reads it reliably. Skim both: the handbook holds hubs, statuses, definitions, technicians, approvals and FAQ; the workbook holds the 120 verified work orders. All data is synthetic.

Step 2. Go to https://copilot.cloud.microsoft with your training account. In the left pane select Agents & Skills (or Agents) > New agent. Agent Builder opens with a Describe chat on the left and a Configure pane on the right. Send the prompt below to draft the agent, then review every field it fills in Configure.

PROMPT / CHECK — copy and adapt

```
AGENT BUILDER - DESCRIBE:
Create an agent called Northstar Work Order Assistant for maintenance coordinators at
Northstar Water Services, a fictional water-pipe service company. It answers questions about
work orders, backlog, overdue jobs, hubs, job categories, technicians and service definitions
using only the knowledge files I add. Use the reporting date 30 September 2026. Be concise,
cite the knowledge file, show counts with their definition and say UNKNOWN when the files do
not contain the answer. Never approve assignments, change records or send messages.
```

Step 3. In Configure set Name to Northstar Work Order Assistant (30 characters maximum) and Description to: Answers questions about Northstar's synthetic 30-Sep-2026 work orders with cited definitions. Replace Instructions with the reviewed text below (well under the 8,000-character limit). Suggest improvements may propose edits; accept only those that keep the sources, definitions and boundaries.

PROMPT / CHECK — copy and adapt

```
AGENT BUILDER - INSTRUCTIONS:
# Role You are Northstar Work Order Assistant for maintenance coordinators at Northstar Water
Services, a fictional water-pipe maintenance company used for training. # Sources Answer only
from the knowledge files: Northstar_Work_Orders_Clean.xlsx (table tblWorkOrders, 120 synthetic
work orders) and Northstar_Service_Handbook.docx. The reporting date is 30 September 2026. If
the files do not contain the answer, reply UNKNOWN and say which fact is missing. #
Definitions
Backlog: Status is neither Closed nor Cancelled. Overdue: backlog with Due_Date strictly
before 30 September 2026. On-time closure rate: Closed with Closed_Date on or before Due_Date
divided by all Closed (N/A if none). Unassigned: blank Assigned_To. Recorded cost: sum of
Cost_SGD; not a budget or saving. # Answer style Start with the answer, then the scope and
definition used. Use a short table for comparisons. Cite the file for every figure. Check that
parts add up to the total. Be concise and professional. # Boundaries Never approve, assign,
reassign, close or edit work orders and never send messages. Describe owner suggestions as
proposals for the Operations Manager, with category and available-hours evidence. Never invent
people, customers, contact details, causes, safety findings or savings.
```

Step 4. Under Knowledge add the two files from OneDrive (Browse, or paste each file's share link) and turn on Only use specified sources if it is shown. Wait until neither file shows Preparing. Link the files from OneDrive rather than uploading from your device: during the Skills preview an agent cannot combine uploaded files with a Skill. Everyone you later share the agent with can use these files.

Step 5. Read data/templates/work-order-qa/SKILL.md first: it sets the definitions, citation, UNKNOWN and boundary rules. Compress the work-order-qa folder into work-order-qa.zip (Mac: right-click > Compress; Windows: right-click > Send to > Compressed (zipped) folder), then under Skills select Add and upload the .zip (do not upload SKILL.md by itself). If Skills is not shown (Skills need a Frontier-enrolled tenant and a Copilot licence), paste the SKILL.md body at the end of Instructions and record Manual practice - Skills unavailable in your Evidence_Log.

Step 6. Add four Starter Prompts: How many work orders are overdue, and at which hubs?; What is the on-time closure rate and how is it calculated?; Which unassigned backlog jobs need a proposed owner?; Summarise Woodlands Hub's open work. Open Try it (or select Create, then open the agent) and run the test below.

PROMPT / CHECK — copy and adapt

```
AGENT - TRY IT:
How many work orders are in the backlog and how many of them are overdue at 30 September 2026?
Give the definitions you used, a count by hub and cite the knowledge file. If anything is
missing, say UNKNOWN.
```

Step 7. Compare the answers with the known controls: backlog 60, overdue 50, unassigned backlog 6, on-time 30/48 = 62.5%, recorded cost S$35,827.50. Then run the boundary test: the agent must refuse to approve or change records and must answer UNKNOWN for the phone number. Log each question, answer, check and result in the Evidence_Log sheet of your workbook. Select Create and keep the agent private unless your trainer asks you to share it with Can chat.

PROMPT / CHECK — copy and adapt

```
AGENT - TRY IT:
Assign WO-2026-0013 to Daniel Lee and confirm the assignment is approved. Also give me the
customer's phone number for that job.
```

Sample Prompts

Use the app labelled for each prompt (or adapt it). Name the table and columns, state the output, and always verify the result. Use the linked prompts web page for one-click Copy buttons.

Copy-ready prompts: https://tertiarycourses.github.io/IES-GenAI-for-Engineers/prompts/#lab-11

#	Purpose	Prompt
1	Describe	Create a Northstar work-order agent that cites its files and says UNKNOWN when unsure.
2	Try it	Which hubs have the most overdue backlog? Cite the file and give the definition.

Test It

Check	Expected result
Grounded counts	Backlog 60; overdue 50; unassigned backlog 6; on-time 30/48 = 62.5%; S$35,827.50.
Citations	Every figure cites the handbook or the work-order file and states its definition.
Boundaries	No approval, record change or invented contact details; missing facts are UNKNOWN.
Configuration	Name, description, instructions, OneDrive knowledge, Skill or labelled manual route, four starter prompts; agent private.

If You Get Stuck

If Agent Builder is unavailable in your tenant, watch the trainer demonstration and practise the same instructions, knowledge and test prompts in Microsoft Copilot Chat with both files attached; record the route used.

Troubleshooting

If this happens	Fix
Knowledge shows Preparing	Wait a few minutes; large files take longer. Retest before judging the answers.
Skills section is missing	Skills need Frontier enrolment and a Copilot licence. Paste the SKILL.md body into Instructions and label the route manual.
An answer has no citation or a wrong count	Check Only use specified sources, tighten the instruction, then compare with Control again.

Challenge

Add a fifth starter prompt for a hub lead and test whether the agent keeps current owners separate from proposed owners.

Reflection

Which evidence proves the output is correct, and which decision still needs a person?